NIKOLAI NEKRUTENKO

nekrutnikolai@gmail.com · nekrutnikolai.com

EDUCATION

Cornell University · Ithaca, NY

Master of Engineering · Electrical and Computer Engineering *Aug 2024 – May 2025*

Relevant Coursework: Microcontrollers, Analog IC Design, Foundations of Robotics, Foundations of Machine Learning, Multi-Agent Systems, Power System and Market Operations.

Bachelor of Arts · Physics *Aug 2021 – Dec 2024*

Relevant Coursework: Electronic Circuits, Embedded Systems, Signals & Systems, Data Science, Radar Remote Sensing, Electro/Magnetostatics & Electrodynamics, Analytical Mechanics.

SKILLS & LICENSES

Programming Languages & Tools: Python, C++, Linux, Bash Scripting, Git, NumPy, SciPy, Observable JS

Embedded & Electrical Systems: Microcontrollers, Sensor Interfacing, Circuit Design, Signal Processing

Prototyping Skills: CAD, KiCad, Electronics Prototyping, SLS/SLA/FDM 3D Printing

Other Skills: Problem Solving, Team Collaboration, Technical Documentation, Project Management

Languages & Licenses: Fluent in Russian, French, English · FAA Part 107 Remote Pilot License, General FCC Radio License

WORK EXPERIENCE

Avionics Hardware Development Engineer · Varda Space Industries · El Segundo, CA *Aug 2025 – Present*

Maintain and develop in-house GNC sensors for the Varda Bus.

Design custom electronics and instrumentation for pharmaceutical manufacturing.

Systems Integration Intern · Formlabs · Somerville, MA *Jun 2024 – Aug 2024*

Devised an experimental setup and methods to evaluate thermopile array infrared (IR) sensors for next-gen SLS printing technology to maximize performance and dimensional accuracy.

Calculated the off-axis projection of sensor pixels and made an interactive tool in Observable JavaScript.

Developed test and alignment scripts in Python, automating the test setup to characterize IR sensors.

Built analysis Jupyter Notebooks in Python to compare the performance of sensors across varying environmental conditions and sensor configurations using NumPy and SciPy.

Gimbals R&D Intern · Freefly Systems · Woodinville, WA *Jun 2023 – Apr 2024*

Developed a motion-control beta embedded firmware release in C++ with Git version control for the Movi Pro gimbal based on IMU data, gimbal physics, and user input — used in cinema productions.

Created an intuitive menu with user feedback on the Movi Controller to interact with the beta firmware.

Wrote user-friendly, interactive documentation outlining the beta firmware in Observable JavaScript.

Derived a forward and inverse kinematics model of a non-orthogonal gimbal to evaluate the viability of a potential gimbal design and determine gimbal-lock mitigation techniques.

Student Researcher · Fatemi Lab @ Cornell AEP · Ithaca, NY *Mar 2022 – Feb 2024*

Wrote automated installation scripts and Jupyter Notebooks to set up a Linux environment for interfacing with lab equipment over the GPIB interface using Python and QCoDeS.

Saved tens of thousands of dollars in lab equipment by enabling interfacing with existing equipment.

Simulated variations in resonance due to EM fields, informing the design of microwave resonators to measure impurities in substrates, in collaboration with grad students — increasing cleanroom efficiency.

Presented project updates at weekly lab meetings and to collaborators.

Teaching Assistant · Cornell University · Ithaca, NY *Aug 2022 – Dec 2022*

- Led a weekly lab section of 15 students and held weekly office hours.
- Graded problem sets and labs, giving constructive feedback to help students understand key ideas.
- Mentored students about future course and career advice.

PROJECT EXPERIENCE

Low-Cost Depth Sensing Module for Deep-Water Research · M.Eng. Thesis *Sep 2024 – May 2025*

Developed a custom sensor mechanism and embedded systems package in collaboration with Dr. Hunter Adams and the Woods Hole Oceanographic Institution for depths up to 2000 meters.

Drone Development Member · Cornell University *Sep 2023 – May 2025*

Developed a robust, low-cost quadcopter platform to be used in mechanical, aerospace, and electrical engineering course labs as a supplementary hands-on lab component with a team of engineers.

NeoPixel FFT Audio Visualizer · Personal Project *Jun 2022 – Jan 2023*

Co-designed and wrote a program that visualizes the waveform and intensity of music for a custom-built individually-addressable RGB LED matrix using Fast Fourier Transforms in Python on a Raspberry Pi.

EXTRACURRICULAR ACTIVITIES

First Degree Black Belt · A Mountain Wind Martial Arts · State College, PA *Feb 2013 – Present*

Instruct, help lead class, and aid students with the practical and philosophical applications of the martial art at a local studio. First Degree Black Belt in Tang Soo Do, a Korean martial art.

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